

2 Goal and Area of Application

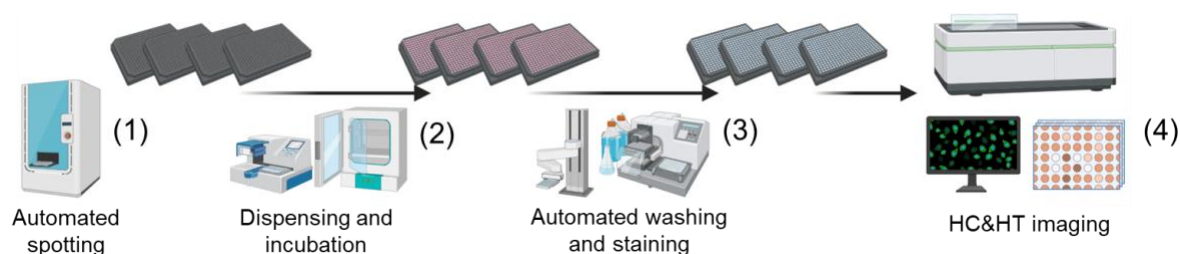
This guideline describes the protocol of using high-content imaging to evaluate small molecule-induced phospholipidosis in cells, which might be an undesirable mechanism of action (MOA) of nuisance compounds in cell-based assays. The protocol has been applied for U2OS and A549 cells but can be adapted to other cell lines.

3 Definitions and Abbreviations

HCS	High-Content Screening
PL	Phospholipidosis
MOA	Mechanism of action
PFA	Polymeric formaldehyde
PBS	Phosphate buffered saline

4 Method

4.1 Procedure



1. Assay-ready plates with pre-spotted compounds were obtained from the Compound Center -20°C freezer and thawed at room temperature for >1 hr prior to removal of seals.
2. U2OS cells cultured in T175 flask were detached with trypsin upon 70-90% confluency and diluted to 2.7×10^5 cells/mL with media. LipidTox green diluted at 1:1000 in media went through a $0.2 \mu\text{m}$ membrane filter and mixed with the cell suspension at 1:1 volume for a final of 1:2000 dilution and 1.35×10^5 cells/mL.
3. The mixture was dispensed by a Multidrop at $30 \mu\text{L}$ and 4000 cells per well to the pre-spotted plates. The plates were stacked, covered with wet paper towels, and stored in a plastic box inside of a cell incubator.
4. After 24 hours, cells were fixed by adding $10 \mu\text{L}$ of 16% PFA to each well and incubated for 20 minutes at room temperature. Then the cells were washed three times with PBS on a plate washer (Aquamax) and incubated with $40 \mu\text{L}$ /well of Hoechst (diluted to $2 \mu\text{g/mL}$) and CellMask deep red (diluted at 1:20000) in PBS for 1 hour at 37°C . The plates were washed again three times and stored in $40 \mu\text{L}$ PBS at 4°C until use.
5. Pre-warm plates at RT for 1 hr. Images were obtained from a fully automated Opera Phenix HCS system.

4.2 Other Sections

Plate layout and controls

Positive controls tamoxifen (TAM), amiodarone (AMI), sertraline (SER) and hydroxychloroquine (HCQ) were spotted in all plates for a final concentration of 10 μ M. HCQ was dissolved in water. DMSO and water were negative controls. See below.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
A	TAM	DMSO																					TAM	DMSO
B	TAM	DMSO																					TAM	DMSO
C	AMI	Water																					AMI	Water
D	SER	TAM																					SER	TAM
E	HCQ	TAM																					HCQ	TAM
F	DMSO	AMI																					DMSO	AMI
G	DMSO	SER																					DMSO	SER
H	Water	HCQ																					Water	HCQ
I	TAM	DMSO																					TAM	DMSO
J	TAM	DMSO																					TAM	DMSO
K	AMI	Water																					AMI	Water
L	SER	TAM																					SER	TAM
M	HCQ	TAM																					HCQ	TAM
N	DMSO	AMI																					DMSO	AMI
O	DMSO	SER																					DMSO	SER
P	Water	HCQ																					Water	HCQ

4.3 Information for consumables/ chemicals/ equipment

Compound handling: The CBCS annotated library (information available at <https://compoundcenter.scilifelab.se/>), which includes SPECS repurposing set and additional sub-libraries with known compounds were used for the PL screen. The compounds were stored in 10 mM DMSO solution and housed at the Compound Center. Four positive controls were provided separately. The empty plates were spotted with 30 nL of compounds each well using Echo 550 (Labcyte), then heat-sealed and stored at -20°C freezer. A total of 25 plates were obtained for the primary screen, 7 plates for confirmation screen and 12 plates for dose-response screen.

Imaging plates: 384-well Pheno-plate, black, optically clear flat-bottom, Tissue Culture treated. Catalog number cat#6057302, Revvity.

Cell culture media: McCoy 5A media (Sigma M8403) with 10% FBS (Gibco 10500064) and 1X of Penicillin-Streptomycin-Glutamine (100X) mixture (Gibco 10378016).

Fluorescent dyes: LipidTox green (Invitrogen #H34350), Hoechst 33342 (Sigma #B2261), and CellMask Deep Red (Invitrogen #H32721).

4.4 Measurements

Measurements and image analysis were performed on the Opera Phenix HCS system.

Measurement settings: the DAPI (blue), GFP (green) and deep red channels were applied to capture signals from Hoechst, LipidTox Green and Cellmask deep red, respectively. Images were taken at four sites in the middle of each well and merged for analysis.

Image analysis and data processing: the nucleus was identified from the blue and the whole cell identified from deep red. Then the cytoplasm region was defined from subtracting whole cells with nucleus. LipidTox green stained spots within the cytoplasm region were further identified. Border objects were excluded. Cell numbers were counted from the nucleus

staining and used to indicate compound toxicity. Total spot number or total area of green per cell in each well were used to evaluate compound induced PL effect. Image analysis results were saved as csv files and subjected to data analysis in excel. Assay quality control was assessed, using tamoxifen as the positive control and DMSO as the negative control, via the BREEZE platform (<https://breeze.fimm.fi/v2/>). Primary hits were selected based on PL induction values above the average+3xstd of all samples (excluding controls), while toxic compounds were identified (cell number <80% of DMSO controls). Hits were confirmed in triplicates and dose-response screenings at non-toxic concentrations.

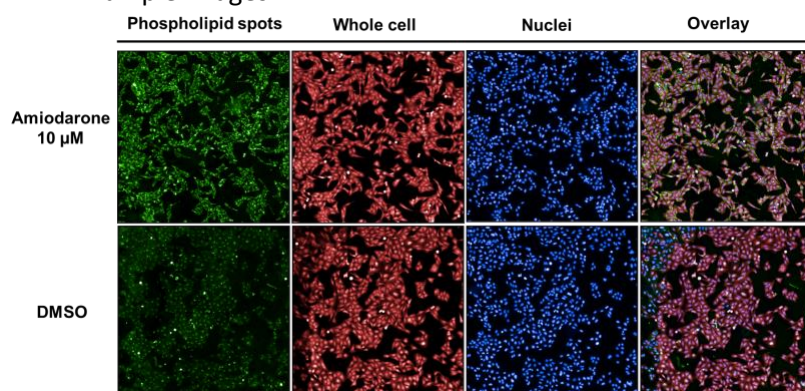
5 Further applicable documents

Literature references

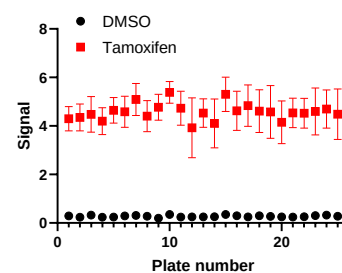
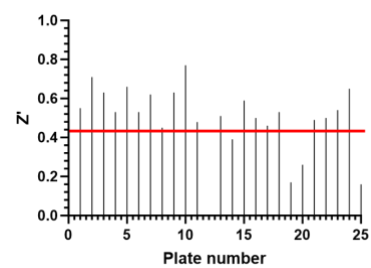
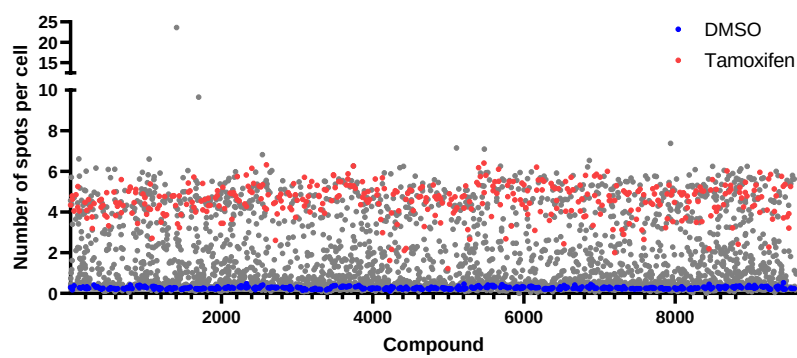
1. Shahane, Sampada A., et al. "Detection of phospholipidosis induction: a cell-based assay in high-throughput and high-content format." *Journal of biomolecular screening* 19.1 (2014): 66-76.
2. Hu, Huabin, et al. "A machine learning and live-cell imaging tool kit uncovers small molecules induced phospholipidosis." *Cell chemical biology* 30.12 (2023): 1634-1651.

6 Annex

Example images



Result analysis



Revision

SOP-Number	Version	Reason for the change	Date	Sign.